Supplementary Material S5. Extended Study-Level Taxonomy of the Included Studies (n = 71)

"No." refers to the record number in S1; "Ref." refers to the reference number in the paper. Limitations and future research directions are as reported by the original authors.

Database	No.	Ref.	Author(s)	Application Area	Prediction Task	Models	XAI method(s)	Performance metrics	Explanation level	Data Type	Model complexity	Explanation output (XAI findings)	Limitations	Future Research Directions
Scopus	72	8	Farooq et al. (2026)	Transportation systems / Vehicle emissions	Fuel consumption and CO_2 emissions prediction	RF, AdaBoost, GB, ETR, XGBoost, LightGBM, LR, Ridge, Lasso, SVR	LIME (Local)	R^2, RMSE, MAE, MSE	Local	Historical structured tabular vehicle dataset (Kaggle, Canadian light-duty vehicles, primarily 2014–2017)	Black-box: RF, AdaBoost, GB, ETR, XGBoost, LightGBM, SVR; Interpretable: LR, Ridge, Lasso	Engine size and cylinders influence fuel consumption, while combined fuel consumption and engine size are key contributors to CO_2 predictions.	Small, geographically restricted dataset; limited real-time validation; XAI computational overhead.	Real-time IoT validation, larger and more diverse datasets, and advanced XAI methods (SHAP/hybrid XAI).
Scopus	70	9	Ranjan and Madanayak (2026)	Transportation / Freight Operations	CO_2 emission intensity prediction (real-time, micro-operational level)	MLR, LASSO, BRR, RF, XGBoost	SHAP (global & local), Partial Dependence Plots	R^2, RMSE, MAE, MSE	Global + Local	Real-world, second-by-second operational and emissions data from 14 port-based HDVs (375,000 observations).	Mixed (white-box: MLR; black-box: RF, XGBoost)	Fuel consumption is a dominant driver of CO_2 emissions, while speed, acceleration, engine displacement, vehicle age, and mileage also influence emission predictions.	Single-port dataset involving 14 HDVs; limited spatial and operational coverage.	Advanced model tuning, integration of spatial and road-network data, hybrid dispersion modeling, and prediction of additional pollutants.
Scopus	110	10	Şahin (2025)	Maritime transportation / Passenger vessels / Carbon emissions	Hourly CO_2 emissions prediction from passenger ships	RF, XGBoost	SHAP (Global)	R^2, RMSE, MAE, Relative MAE (%)	Global	Passenger vessel structural and operational data	Black-box: RF, XGBoost	Gross tonnage (GT), speed, and length identified as key drivers	Passenger-only data, fixed emission factors, and limited operational context	Broader vessel types, diverse operating conditions, and real-time AIS integration
Scopus	42	11	Özüpak (2026)	Aviation / Air transport sustainability	Aviation CO_2 emissions prediction and high-emission detection	Physics-based baseline, LSTM, 1D-CNN, Transformer, Hybrid residual model, Stacked Ensemble (Ridge meta-learner)	SHAP (Global + Local), LIME (Local)	RMSE, MAE, R^2, AUC, NSE, Theil U2	Global + Local	Structured time-series aviation emissions and traffic data	Hybrid black-box	Traffic and lagged CO_2 were the main drivers.	Europe-only, aggregated data.	Multi-region expansion; flight-level and real-time data.
Scopus	50	12	Gungor et al. (2026)	Building energy performance / Campus CO_2 emissions	Building-level annual CO_2 emissions prediction	SVR, SVR_BO, GPR, LSBoost, LSBoost_PI, ANN, Bagged Trees, Regression Tree, Ridge Regression, Stacking Ensemble	SHAP (Global + Local), Permutation Importance (Global), Ablation Analysis (Global)	RMSE, MSE, MAE, MedAE, Bias, R^2, MAPE, NRMSE, EVS, Corr, RMSLE	Global + Local	Structured building-level energy and emissions data (18 campus buildings)	Mixed (White-box: Ridge Regression, Regression Tree; Black-box: SVR, SVR_BO, GPR, LSBoost, LSBoost_PI, ANN, Bagged Trees, Stacking Ensemble)	CO_2/kWh and CO_2/TEP dominate predictions	Small, single-campus dataset	Multi-campus and cross-climate validation
Scopus	32	13	Wani et al. (2026)	Construction materials / Sustainable concrete	CO_2 emissions prediction and minimization	XGBoost (XGB), LightGBM (LGBM), CatBoost (CB), AdaBoost (AB), Neural Network (NN)	SHAP (Global + Local), LIME, PDP	R^2, MAE, RMSE, MSE, NSE, MAPE	Global + Local	Structured mix-design data	Black-box: CB, XGB, LGBM, AB, NN	GGBFS content, activator ratio, and curing temperature were the main CO_2 drivers.	Small dataset (n=131) and limited generalizability.	Larger datasets, regional emission factors, and multi-objective optimization.
Scopus	16	14	Zhu et al. (2026)	Urban carbon emissions / Nighttime-light-based estimation / Regional mitigation	City-level carbon emission prediction and heterogeneous driver analysis in the Yangtze River Delta	DT, RF, XGBoost, SVM, KNN	SHAP (Global + Local), PDP (Global + Interaction)	R^2, MSE, MAE	Global + Local + Interaction	City-level panel data; 41 cities, 2005–2022; 738 observations; seven selected drivers; nighttime-light-derived emissions and socioeconomic statistics	Interpretable: DT; Black-box: RF, XGBoost, SVM, KNN	Dominant drivers: population, technological innovation, openness, and per capita GDP; nonlinear effects, population–innovation interaction, and zonal differences	City-level resolution; limited interaction quantification; aggregated innovation indicator	County-level remote-sensing integration; SHAP interaction analysis and multidimensional PDPs; green-patent indicators
Scopus	23	15	Song et al. (2026)	Agriculture / Dairy cattle slurry storage and manure management	Daily CH_4 flux prediction during dairy cattle slurry storage	Linear Regression (LR), Random Forest (RF), XGBoost (XGB), LightGBM (LGBM), Support Vector Regression (SVR), Multilayer Perceptron (MLP)	SHAP (Global + Local)	R^2, RMSE, MAE	Global + Local	Experimental physicochemical, operational, temperature, and temporal data	Mixed (white-box: LR; black-box: RF, XGB, LGBM, SVR, MLP)	Key CH_4 drivers and nonlinear relationships identified	Limited transferability across storage systems and climates	Broader farm-scale validation under diverse storage and management conditions
Scopus	29	16	Song et al. (2026)	Power systems / Substation carbon management	Substation carbon-emission factor (CEF) prediction	RF, GBM, LSTM, Pure Transformer, MOIRAI-2 + UPINN	Any-Variate Attention (Global + Local), UPINN Physical Decomposition (Global + Local), Counterfactual Reasoning (Local)	MAE, RMSE, R^2	Global + Local	Multivariate substation time-series data	Black-box: RF, GBM, LSTM, Transformer; Intrinsically interpretable: MOIRAI-2 + UPINN	Transformer losses, temperature, load factor, and SF_6 leakage explain CEF dynamics.	Single-station, 365-day dataset.	Multi-station validation, longer-term data, and probabilistic forecasting.

Database	No.	Ref.	Author(s)	Application Area	Prediction Task	Models	XAI method(s)	Performance metrics	Explanation level	Data Type	Model complexity	Explanation output (XAI findings)	Limitations	Future Research Directions
Scopus	192	17	Assael et al. (2023)	Corporate sustainability / GHG emissions / Carbon accounting	Estimation of non-reported corporate Scope 1 and Scope 2 GHG emissions	LightGBM	SHAP (Global + Local)	R^2, RMSE, MAE	Global + Local	Corporate financial, energy, industry, regional, and Scope 1–2 GHG emissions data (2010–2020)	Black-box: LightGBM	SHAP identified energy consumption and industry classification as major predictors	Incomplete disclosures, limited SME data, and industry classification limitations	Expanded SME data, finer industry classification, and SHAP-based data refinement
Scopus	108	18	Zhou et al. (2025)	Carbon markets / Carbon pricing / Investment decision support	One- and three-step-ahead carbon price point and interval forecasting	DNMADE, GS–ESN, GS–BPnn, GS–LSTM, GS–CNNLSTM, GS–Transformer, GS–DLinear, MGOA–ORELM, CWOA–ELM, SSA–BPnn, SMA–BiLSTM, HKDE, LUBE, QTCN, GPR	AW (Global + Local), SHAP (Global + Local)	MAE, RMSE, MAPE, MSPE, IA; PICP, PINAW, AIS, CWC; DA	Global + Local	Daily prices and multi-source predictors from EU, New Zealand, Hubei, and China's national ETS; market-specific periods within 2013–2025	Linear: GS–DLinear; Black-box neural models: DNMADE's AttESN, GS–ESN, GS–BPnn, GS–LSTM, GS–CNNLSTM, GS–Transformer, MGOA–ORELM, CWOA–ELM, SSA–BPnn, SMA–BiLSTM, LUBE, QTCN; Nonparametric statistical models: HKDE, GPR	Predictor contributions varied across markets, lags, trends, and residuals; sentiment and uncertainty were more prominent in short-term fluctuations	Limited representation of policy texts and public opinion; scope for richer forecasting and explanation methods	Policy-text and sentiment integration; advanced trend forecasting; improved interpretability methods
Scopus	158	19	Liang et al. (2025)	Carbon markets / Emissions trading / Carbon pricing	Carbon price forecasting in China's Hubei carbon market	LR, LGBM, ANN, KNN, LSTM, ARIMA, EEMD–GAMI_NET–LSTM–ARIMA	ALE (Global)	R^2, RMSE, MAE, DA	Global	Historical market and economic time-series data from Wind; 892 observations; 16 input features; January 2020–November 2023	Mixed: Mixed: EEMD–GAMI_NET–LSTM–ARIMA; (Black-box: LGBM, ANN, KNN, LSTM; White-box: LR, ARIMA; Interpretable component: GAMI_NET; White-box component: ARIMA; Black-box component: LSTM; Decomposition method: EEMD)	Key drivers: lagged prices, trading volume, energy prices, exchange rates, and international carbon prices	Single market; limited period; over-decomposition reduced accuracy	Cross-market validation; refined decomposition; early-warning applications
Scopus	4	28	Ye et al. (2026)	Transportation / Urban carbon emissions	Transportation CO_2 emissions prediction	LR, RF, XGBoost, LightGBM, CatBoost	SHAP (Global + Local)	R^2, RMSE, MAE, MAPE	Global + Local	Structured city-level transport and socioeconomic panel data	Black-box: RF, XGBoost, LightGBM, CatBoost	Socioeconomic factors dominate; higher compactness is associated with lower predicted TCE; freight gravity increases predicted emissions	YRD-specific data; EV-related upstream emissions excluded; historical-data dependence	Broader geographic validation; include upstream EV emissions; adaptive AI for unforeseen changes
Scopus	2	29	Özüpak and Aslan (2027)	Greenhouse gas emissions / Climate change	Long-term CO_2 and N_2O emissions prediction	DT, KNN, AdaBoost, SVM, Optimized Random Forest (RF)	SHAP (Global + Local), LIME (Local)	R^2, RMSE, MAE, MSE, Max Error	Global + Local	Structured country-level panel data (GHG inventories and socio-economic indicators)	Black-box: RF, AdaBoost, SVM	GDP dominates both CO_2 and N_2O predictions; population also contributes to N_2O emissions	Outdated data; limited predictors; unvalidated long-term projections.	Updated inventories; sectoral variables; multi-scenario forecasting; uncertainty quantification
Scopus	41	30	Yang et al. (2026)	Carbon markets / Regional ETS / Carbon pricing	Daily carbon price forecasting across China's eight ETS pilots	XCPFNet (CEEMDAN–CNN–LSTM–Transformer), EMD–TI–LSTM, CEEMDAN–LSTM–Transformer, CEEMDAN–SVR–LSTM	Clustered SHAP (Global + Local + Interaction); Kernel SHAP (Global + Local; comparison baseline)	MSE, MAE, RMSE, MAPE, R^2; PIPC, PINAW; ES, FIC, TS	Global + Local + Interaction	Daily prices from eight Chinese ETS pilots; 2292 trading days, April 2014–August 2023; 12 driving variables	Black-box hybrid models: XCPFNet, EMD–TI–LSTM, CEEMDAN–LSTM–Transformer, CEEMDAN–SVR–LSTM	Energy prices and EUA were major contributors; policy-uncertainty effects varied across regions; nonlinear interactions and attribution stability were assessed	Correlation-based explanations; daily data resolution; regional-pilot coverage	Causal and counterfactual analysis; high-frequency market data; national–international market integration
Scopus	1	31	Ali et al. (2027)	Agriculture / Agri-food systems / Carbon management	CO_2 emission prediction	RF	SHAP (Global), LIME (Local), DiCE (Counterfactual)	R^2, MSE, RMSE, MAE, MAPE, Median Absolute Error, Max Error, Explained Variance Score	Global + Local + Counterfactual	Structured agri-food environmental data	Black-box: RF	Food Processing ranked first; Net Forest Conversion second (permutation importance).	Sector-specific data; sensitivity to data quality; theoretical conversion assumptions; limited real-world feasibility; rule-based AI	Cross-sector validation; multimodal data; techno-economic analysis; adaptive AI; digital twins

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Scopus	8	32	Saraswat et al. (2026)	Transportation systems / Vehicle emissions	Vehicle CO_2 emissions prediction	MLP, GAO-MLP (MLGO), HLOA-MLP (MLHO), XGBoost (XGB), Random Forest (RF), SVR	SHAP (Global + Local), RFE, CAM	RMSE, R^2, MAPE, MARE, TIC	Global + Local	Structured vehicle attributes and fuel-consumption data	Black-box: MLP, XGB, RF, SVR	Combined fuel consumption and engine size dominate CO_2 predictions	Static data and missing powertrain features	Real-time telemetry and physics-informed models
Scopus	27	33	Shaik et al. (2026)	Transportation systems / Vehicle emissions	Vehicular CO_2 emissions prediction	HEO-TabNet, PSO-TabNet, FFO-TabNet	TabNet Feature Masks (Global + Local), SHAP (Global + Local), LIME (Local)	RMSE, MAE, R^2, MAPE	Global + Local	Structured vehicle specifications and fuel-consumption data	Black-box: TabNet-based optimized DL models	Fuel consumption and engine size identified as key predictors	No external validation; limited model benchmarking	External validation and broader model benchmarking
Scopus	90	34	Saleem et al. (2026)	Transportation / Connected and autonomous vehicles (CAVs)	Fuel consumption and CO_2 emission prediction	Extra Trees, Linear Regression, XGBoost, Bayesian Ridge, Ridge Regression, Gradient Boosting, RF, DT, AdaBoost, LASSO, SVR, KNN, ElasticNet, LightGBM, Huber Regressor	LIME (Local), SHAP (Global)	R^2, RMSE, MAE	Global + Local	Vehicle-level tabular data	Black-box + Interpretable	Engine size and cylinders identified as key drivers	Simulated FL, single dataset, and limited validation	Real-world FL validation, diverse datasets, and enhanced privacy
Scopus	109	35	Alam et al. (2025)	Transportation / Vehicle emissions / Environmental sustainability	Vehicle CO_2 emission prediction	DT, KNN, SVR, XGBoost, AdaBoost, CatBoost, Ensemble (DT + KNN + XGBoost), LSTM, BiLSTM, CarbonMLP	SHAP (Global + Local)	R^2, MSE, RMSE, MAPE	Global + Local	Canadian vehicle-level tabular emissions data	Black-box: DT, KNN, SVR, XGBoost, AdaBoost, CatBoost, Ensemble, LSTM, BiLSTM, CarbonMLP	Fuel consumption and engine characteristics identified as key emission drivers	Canada-specific dataset and lack of real-world validation	Broader datasets, additional pollutants, and real-world validation
Scopus	131	36	Yuan et al. (2025)	Transportation / Vehicle emissions / Sustainable transport	ICE and HEV CO_2 emissions prediction	LR, DT, RF, XGBoost	ALE (Global), Feature Importance Analysis (Global)	R^2, MSE, RMSE	Global	UK vehicle certification and emissions data	Interpretable: LR, DT; Black-box: RF, XGBoost	Fuel consumption, engine capacity, and noise-related thresholds identified	Missing engine boosting parameters; certification-based data	Incorporation of direct engine boosting parameters
Scopus	167	37	Guo et al. (2024)	Transportation / Vehicle emissions	Vehicle CO_2 emission prediction	KNN, SVR, GBM, DT, RF, LASSO	SHAP (Global), ALE (Global)	R^2, RMSE, rRMSE, MAE, MAPE	Global	Real-world Canadian vehicle specifications and fuel consumption data	Black-box: KNN, SVR, GBM, DT, RF; Interpretable: LASSO	Key CO_2 emission drivers and feature effects identified	Canada-only vehicle data; limited generalizability	Multi-region and multi-sector data integration
Scopus	33	38	Luo et al. (2026)	Transportation systems / Tunnel CO_2 emissions	Instantaneous vehicle CO_2 emissions prediction	VSP, MLR, RF, XGBoost	SHAP (Global), ICE (Local)	R^2, RMSE	Global + Local	High-resolution PEMS vehicle data	Mixed (white-box: VSP, MLR; black-box: RF, XGBoost)	Engine load, engine speed, VSP, and exhaust temperature identified as key CO_2 drivers	Single-vehicle pilot data	Broader PEMS validation
Scopus	28	39	Chen et al. (2026)	Transportation systems / Transportation carbon emissions	Transportation CO_2 emissions forecasting	Extra Trees (ET)	SHAP (Global + Local)	MSE, RMSE, R^2	Global + Local	Structured city-level transportation, urbanization, socioeconomic, and energy data	Black-box: ET	Transport service output, vehicle ownership, and energy consumption identified as key drivers	Macro-level analysis, limited indicators, and generalized scenarios	Micro-level simulations, richer indicators, and policy impact assessment
Scopus	116	40	Huang et al. (2025)	Urban transportation / Traffic carbon emissions / Sustainable urban planning	Neighborhood-level road traffic CO_2 emission prediction	XGBoost	SHAP (Global + Local), GAM-based SHAP dependence analysis (Global)	RMSE, MAE, R^2	Global + Local	Geospatial built-environment, traffic, POI, and CO_2 emission data	Black-box: XGBoost; Interpretable: GAM	Key built-environment drivers and nonlinear thresholds identified	Three measurement areas, missing weather data, and single-city scope	Broader spatial validation, meteorological variables, and interaction analysis
Scopus	133	41	Zhao and He (2025)	Urban transportation / Carbon emissions / Sustainable mobility	City-level transportation carbon emission prediction	CatBoost, LightGBM, XGBoost, RF, GBR, MLP, ELM, Bayes, KNN, DT	SHAP (Global + Local + Interaction)	R^2, RMSE, MAE, MAPE	Global + Local + Interaction	City-level transportation emissions and socioeconomic panel data (2000–2021)	Interpretable: Bayes, DT; Black-box: CatBoost, LightGBM, XGBoost, RF, GBR, MLP, ELM, KNN	Built-up area, fiscal revenue, and nonlinear feature interactions identified	Limited causal inference, coarse data granularity, and model selection bias	Causal inference methods, finer-resolution data, and multi-model validation
Scopus	137	42	Roy et al. (2025)	Urban transportation / Traffic emissions / Sustainable mobility	Joint prediction of intersection-level CO_2 emissions and travel delay	SVR, RF, XGBoost, GB, MTCNN, MT2CN	SHAP (Global + Local + Interaction)	R^2, RMSE, MAE, MAPE	Global + Local + Interaction	Simulated spatiotemporal traffic, signal timing and vehicle data	Black-box: SVR, RF, XGBoost, GB, MTCNN, MT2CN	Signal timing, traffic demand, and HDV share identified as key drivers	Single intersection and limited signal phase configurations	Multiple intersections and diverse signal phase configurations

Database	No.	Ref.	Author(s)	Application Area	Prediction Task	Models	XAI method(s)	Performance metrics	Explanation level	Data Type	Model complexity	Explanation output (XAI findings)	Limitations	Future Research Directions
WoS only	70	43	Liu et al. (2025)	Urban transportation / Metro–bus combined travel	Metro-bus combined travel (MBCT) CO_2 emission	GBDT, XGBoost, LightGBM, CatBoost, OLS	SHAP (Global + Local), PDP (Global)	R^2, RMSE, MAE	Global + Local	Smart-card transactions, bus operation records, and built-environment data (8,149 metro–bus combined trips; Shanghai, China; 1 April 2015)	Mixed (Black-box:GBDT, XGBoost, LightGBM, CatBoost; White-box: OLS)	SHAP identified metro distance proportion and car- and metro-route non-linearity coefficients as the strongest predictors; PDP revealed nonlinear effects and thresholds	Symmetric travel-chain assumptions exclude asymmetric trips; single-city and single-day data; comparison restricted to car travel	Integration of GPS trajectories and travel surveys; inclusion of additional transport modes; validation across multiple cities and days
Scopus	73	44	Fang et al. (2026)	Residential buildings / Operational carbon emissions	Hourly operational carbon emission prediction	AR, DT, GBDT, SVM, FNN, GPR, Sageformer, TPA-LSTM, MT-GNN, DCG-TCN-GCN	SHAP (Global + Local)	Corr, RSE, RMSE	Global + Local	Residential building sensor time-series data (279 households, UK)	Mixed (Statistical: AR; Black-box ML: DT, GBDT, SVM, GPR; Black-box DL: FNN, Sageformer, TPA-LSTM, MT-GNN, DCG-TCN-GCN).	Indoor temperature, indoor humidity, and outdoor temperature are key emission predictors.	Single regional dataset; limited generalizability.	Extend to diverse building types and larger geographical scales.
Scopus	114	45	Zhang et al. (2025)	Campus / Buildings / Community-scale carbon management	CO_2 emissions prediction and optimization for a university campus	RF	SHAP (Global), Pearson Correlation Analysis (Global)	R^2, RMSE	Global	Campus energy, demographic, transportation, food, waste, and environmental data	Black-box: RF	Food consumption and energy use identified as key emission drivers	Single-campus data and fixed emission factors	Multi-campus validation and dynamic emission factor integration
Scopus	193	46	Zhang et al. (2023)	Buildings / Building energy efficiency / GHG emissions	Residential building energy consumption and GHG emission prediction	LightGBM, XGBoost, RF, SVR	SHAP (Global + Local), PDP (Global)	R^2, MSE, RMSE, MAE	Global + Local	Seattle residential building energy, GHG emissions, geometry, and urban morphology data (1,523 records)	Black-box: LightGBM, XGBoost, RF, SVR	SHAP identified total GFA and natural gas consumption as the leading predictors of energy use and GHG emissions, respectively	Residential-only data; lack of explicit optimization strategies	Multi-objective optimization; extension to other building types
WoS only	75	47	Zhou et al. (2024)	Building energy efficiency / Office buildings / Carbon emissions	Building carbon emission, PV carbon offset, and net carbon emission prediction	XGBoost, LightGBM	SHAP (Global + Local + Interaction)	R^2, MAPE, MAE	Global + Local	Simulation-based office building energy and carbon data; 24 energy indicators; 1,000 samples	Black-box: XGBoost, LightGBM	SHAP highlighted WWR for building emissions and PVArea/PVEff for PV carbon offset, with nonlinear interactions	Single climate zone; operational carbon emissions only	Validation across climate zones; whole-life-cycle carbon assessment
Scopus	180	48	Khan et al. (2024)	Green buildings / Building energy efficiency / Carbon emissions	Building energy consumption, CO_2 emissions, and thermal discomfort prediction	LGBM, BO-LGBM,BO-XGB	LIME (Local)	R^2, MAPE	Local	BIM-based building energy simulation data	Black-box: LGBM, BO-LGBM, BO-XGB	Key HVAC drivers identified, particularly heating and cooling set-point temperatures	Single-building simulation case study; limited real-world validation	Real-world validation, diverse building types, generative design integration
Scopus	61	49	Cao et al. (2026)	Construction materials / Steam-cured concrete	Construction materials / Steam-cured concrete	RF, BO-RF, XGBoost, BO-XGBoost, CatBoost, BO-CatBoost; improved NSGA-III (optimization)	SHAP (Global + Local)	R^2, RMSE, MAE	Global + Local	Experimental + calculated structured dataset (200 samples)	Black-box: RF, BO-RF, XGBoost, BO-XGBoost, CatBoost, BO-CatBoost; Improved NSGA-III (multi-objective optimization)	Thermostatic temperature, heating rate, and curing duration are the most influential factors on performance and emissions.	Limited to a specific experimental dataset and steam-curing conditions.	Incorporate additional curing and production factors to develop a more comprehensive precast tunnel segment production system.
Scopus	59	50	Wang et al. (2026)	Construction / Prefabricated buildings / Carbon emissions	Carbon emission prediction at component level (regression)	LR, RF, BPNN, XGB	SHAP (global & local), PDP (feature effect analysis)	R^2, MAE, RMSE, MAPE	Global + Local	Structured tabular industrial data (311 prefabricated components from one factory)	Mixed (white-box: LR; black-box: RF, XGB, BPNN)	Volume increases emissions, while higher CSR reduces emissions, particularly above 90%, making it a key controllable design factor.	Single-factory dataset; limited generalizability and no dynamic mold reuse modeling	Validate across factories and incorporate dynamic scheduling and real-time mold reuse data.
Scopus	51	51	Seo et al. (2026)	Construction / Built Environment (Construction equipment emissions)	Carbon emission prediction from construction equipment at pre-construction stage (Regression, Scope 3 emissions)	DT, RF, XGB, SVR, MLP, Ensemble models (DT + SVR)	SHAP (Global + Local + Interaction)	MAE, SMAPE	Global + Local	Structured project-level construction and geotechnical data (96 residential projects)	Mixed (White-box: Decision Tree, Black-box: RF, XGB, SVR, MLP, Hybrid: Ensemble models)	Rippable rock thickness, rock strength, and site area identified as dominant emission drivers	Small, Korea-specific dataset; inter-contractor data variability	Broader project types and international datasets; contractor-specific models
Scopus	88	52	Zheng and Shuang (2026)	Building sector / Embodied carbon emissions	Embodied carbon emission (BEC) prediction	SVM, ANN, RF, XGBoost, CatBoost, BP, LSTM	SHAP (Global + Local)	MAE, MAPE, RMSE, R^2	Global + Local	National-level time-series data	Black-box: SVM, ANN, RF, XGBoost, CatBoost, BP, LSTM	Key carbon emission drivers identified	Limited policy factors and macro-level data	More micro-level indicators and diversified scenarios
Scopus	98	53	Zawaideh (2026)	High-rise construction / BIM-based risk management / Embodied carbon	Cost overrun (%), schedule delay (days), and embodied-carbon intensity ($kgCO_2eq/m^2$) prediction	XGBoost, RF, LightGBM, CatBoost, SVR, ANN	SHAP (Global)	R^2, RMSE, MAE, MAPE	Global	BIM/IFC-derived tabular data; 2,860 scenarios based on 12 high-rise building models (20–60 floors); geometric, material, scheduling, cost, resource, and carbon features	Black-box: XGBoost, RF, LightGBM, CatBoost, SVR, ANN	Cost: budget, concrete, design changes; schedule: critical path, dependencies, MEP clashes; carbon: concrete, steel, transport, emission factors	BIM data-quality dependence; scenario-related overfitting risk; no site-level IoT data	Uncertainty analysis; diverse real-time datasets; site-monitoring integration

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Scopus	115	54	Liu et al. (2025)	Civil engineering / Bridge construction / Carbon emissions	Bridge construction CO_2 emission prediction at the design stage	RF, GBRT, CatBoost, XGBoost, AdaBoost, MLP, DT, SVR	SHAP (Global + Local), PDP (Global), ICE (Local)	MSE, MAE, RMSE, R^2, CWI	Global + Local	Bridge design parameters and LCA-based CO_2 emission data	Black-box: RF, GBRT, CatBoost, XGBoost, AdaBoost, MLP, SVR; Interpretable: DT	Girder volume, bridge length, and pile foundation volume identified as key drivers	Limited sample size and bridge types	Larger, more diverse datasets and broader validation
Scopus	17	55	Chen et al. (2026)	Urban systems / Carbon emissions	Urban carbon emissions prediction	SVR, Random Forest (RF), LightGBM, AdaBoost, CatBoost, XGBoost (XGB)	SHAP (Global + Local + Interaction)	R^2, MSE, RMSE, MAE	Global + Local	Multisource urban spatial and socioeconomic data	Black-box: SVR, RF, LightGBM, AdaBoost, CatBoost, XGB	Key emission predictors, response thresholds, and interactions identified	Region-specific data, limited spatial granularity, and lack of causal inference	Finer-scale analysis, external validation, and causal modeling
Scopus	94	56	Ren et al. (2026)	Urban sustainability / Urban morphology	Sectoral and total carbon emission prediction	ResNet-18 (CNN), XGBoost, Mediation–Moderation Regression	Grad-CAM (Spatial), SHAP (Global + Local)	Accuracy, Precision, R^2, MAE, RMSE, EVS	Spatial + Global + Local	Remote sensing imagery and spatial tabular data	Black-box: ResNet-18, XGBoost; Interpretable: Mediation–Moderation Regression	Key morphological drivers and nonlinear thresholds identified	Limited temporal coverage and spatial validation	Temporal modeling, cross-regional validation, and scenario analysis
Scopus	95	57	Bhowmik et al. (2026)	Forest monitoring / Climate mitigation / Land-use change	Tree cover loss and CO_2 emissions forecasting	LR, RF, XGBoost, LSTM	SHAP (Global)	R^2, RMSE, MAE	Global	State-level spatiotemporal forest data	Black-box: RF, XGBoost, LSTM; Interpretable: LR	Lagged forest loss, emissions, and biomass identified as key predictors	Limited temporal resolution and emission estimation uncertainty	Higher-resolution data and additional environmental covariates
Scopus	99	58	Jiang et al. (2026)	Land-use change / Carbon emissions / Regional sustainability	County-level land-use carbon emission forecasting	XGBoost, RF, PLUS, CA–Markov, InVEST, Carbon Emission Coefficient Method	SHAP (Global + Local + Interaction)	R^2, Kappa	Global + Local + Interaction	County-level spatiotemporal land-use and socioeconomic data	Black-box: RF, XGBoost; Simulation: PLUS, InVEST	Key land-use drivers and nonlinear thresholds identified	Short time series, single-scenario analysis, and carbon accounting uncertainties	Longer time series, multi-scenario analysis, and refined carbon accounting
Scopus	130	59	Bilotta et al. (2025)	Urban sustainability / Smart cities / Carbon emissions	Fine-scale urban CO_2 emission prediction	LR, OLS, XGBoost, GCN	SHAP (Global + Local)	R^2, MAE, MSE, RMSE, MAPE	Global + Local	Urban open data and ODIAC-based CO_2 emissions	Interpretable: LR, OLS; Black-box: XGBoost, GCN	Socioeconomic indicators, fuel stations, and GPS coordinates identified as key drivers	Limited cross-city generalization; lack of online learning and domain adaptation	Domain adaptation, multi-task learning, and spatiotemporal hybrid models
Scopus	132	60	Zhang et al. (2025)	Urban sustainability / Regional carbon emissions / Spatial analysis	Fine-scale energy-related CO_2 emission estimation	LR, SVR, RF, XGBoost	SHAP (Global + Local + Interaction)	R^2, RMSE, MAE, PRD	Global + Local + Interaction	Multi-source satellite, geospatial, and socioeconomic data	Interpretable: LR; Black-box: SVR, RF, XGBoost	GDP and population identified as key drivers; nonlinear thresholds and interactions	Non-locally calibrated emission factors; limited 1 km spatial resolution	Higher-resolution (100–500 m) models integrating POI, imagery, and traffic flow data
Scopus	152	61	Xu et al. (2025)	Urban planning / Residential land carbon emissions	Residential land CO_2 emission prediction and nonlinear relationship analysis	XGBoost, GBDT, RF, AdaBoost, Logistic Regression	SHAP (Global + Local)	R^2, MAE, MSE, RMSE, MAD, MAPE, EVS, MSLE	Global + Local	Residential land, electricity consumption, and built-environment data	Black-box: XGBoost, GBDT, RF, AdaBoost; Interpretable: Logistic Regression	Key emission drivers, nonlinear thresholds, and interaction effects identified	Limited variables, single-region data, and lack of policy simulations	Expand variables, compare models, examine multiple regions and scales, and simulate policies
Scopus	155	62	Wang et al. (2025)	Urban sustainability / Digital transformation	Carbon emission intensity (CEI) prediction	OLS Regression, Ridge Regression, LASSO Regression, GBDT, RF, XGBoost, LGBM	SHAP (Global + Local + Interaction)	R^2, RMSE, MAE	Global + Local + Interaction	Chinese city-level panel data (267 cities, 2010–2022)	Black-box: GBDT, RF, XGBoost, LGBM; Interpretable: OLS, Ridge, LASSO	Key UDT drivers, nonlinear thresholds, and interaction effects identified	Lack of qualitative and spatiotemporal analysis	Incorporate qualitative and spatiotemporal analyses
Scopus	185	63	Luo et al. (2024)	Land use / Regional sustainability / Carbon emissions	Multi-scale CO_2 emission prediction based on land-use changes	LR, MLP, KNN, SVM, ETR, RF, DT, XGBoost, Optimized ETR	SHAP (Global + Local)	R^2, MSE, RMSE, MRE, MAE, MBE	Global + Local	Land-use GIS, remote sensing, POI, and carbon emission panel data	Black-box: MLP, KNN, SVM, ETR, RF, XGBoost; interpretable baseline: LR, DT	Land-use effects and nonlinear emission thresholds identified	Single-region scope; limited temporal coverage and data granularity	Broader regions, longer time series, and finer land-use and emission classifications
Scopus	191	64	Yao et al. (2023)	Urban carbon emissions / Urban sustainability	CO_2 emission turning-point forecasting and identification of socioeconomic drivers	EKC, ARIMA, ECM, DT, SR, XGBoost, GA-EKC	SHAP (Global + Local)	R^2, Adjusted R^2	Global + Local	Beijing socioeconomic, energy, environmental, and CO_2 emission time-series data (1978–2021)	Black-box: XGBoost; interpretable: EKC, ARIMA, ECM, DT, SR; optimization-based: GA-EKC	SHAP identified total energy consumption, life expectancy, and GDP as key predictors	Data limitations and omitted socioeconomic predictors	Additional predictors; advanced feature selection
Scopus	5	65	Sireesha et al. (2026)	Energy / Industrial GHG emissions	Multi-gas GHG emissions prediction and forecasting	GB, RF, XGBoost, CatBoost, AdaBoost	SHAP (Global + Local), LIME (Local), ICE (Local), PDP (Global)	R^2, RMSE, MAE, VAF	Global + Local	Structured historical GHG time-series data	Black-box: GB, RF, XGBoost, CatBoost, AdaBoost	CO_2 dominates predictions; CH_4 and F-gases exhibit sector-specific effects	Historical data uncertainty; limited regional applicability	Real-time data; hybrid and spatio-temporal models
Scopus	74	66	Dahmani et al. (2026)	Climate forecasting / Environmental systems	CO_2 emissions and temperature prediction	MTL-Conv1D-LSTM	LIME (Local)	R^2, MAE, MSE, RMSE	Local	Annual country-level climate, agricultural, land-use, and demographic time-series data (31 variables)	Black-box: Conv1D-LSTM	Recent emission trends dominate CO_2 predictions, while temperature history drives temperature forecasts; forest conversion also influences CO_2 predictions.	Limited atmospheric variables, centralized dataset, untested regional generalizability, and local-only XAI.	Integrate atmospheric and regional data, federated learning, SHAP and counterfactual XAI, and real-time forecasting

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Scopus	76	67	Ghannam (2026)	Environmental Sustainability / National Carbon Emission Forecasting (Australia)	National CO_2 emissions forecasting in Australia	ARIMA, RF, XGBoost, SVR, GPR, ELM, ISSA-ELM, MLP, LSTM, RNN, GRU, Stacking Regressor, Enhanced Stacking Regressor	SHAP (Global), Tree-based feature importance (Global)	R^2, RMSE, MAE, MSE, MAPE, MSLE, MedAE	Global	Annual national economic, energy, and environmental data (1982–2022).	Mixed (white-box: ARIMA, GPR; black-box: RF, XGBoost, SVR, ELM, ISSA-ELM, MLP, LSTM, RNN, GRU, Stacking, Enhanced Stacking)	Energy productivity, GHG emissions, energy consumption, and population identified as key drivers.	Low temporal resolution, national-level aggregation, and omitted policy and trade factors.	Higher-resolution data, policy and trade variables, causal inference, and probabilistic forecasting.
Scopus	87	68	Ma et al. (2026)	Climate change / Carbon emissions	CO_2 emission prediction	RF, XGBoost, LightGBM, CatBoost, MLP	SHAP (Global + Local)	R^2, MSE, RMSE, MAE, MAPE	Global + Local	Multi-country panel data	Black-box: RF, XGBoost, LightGBM, CatBoost, MLP	Key energy-related emission drivers identified	High MAPE sensitivity and simplified policy scenarios	Model optimization and additional policy scenarios
Scopus	151	69	Zheng and Shuang (2025)	Climate / Country-level carbon emissions	Carbon emission, peaking and neutrality prediction	SVM, ANN, RF, XGBoost, CatBoost, BP, LSTM	SHAP (Global + Local)	MAE, RMSE, MAPE, R^2	Global + Local	National-level time-series data	Black-box: SVM, ANN, RF, XGBoost, CatBoost, BP, LSTM	Key socioeconomic and energy-related emission drivers identified	Limited factors, omitted policy effects, and national-level data	Incorporate policy effects, micro-level indicators, diverse metrics, and scenarios
Scopus	153	70	Han and Lin (2025)	Environmental economics / Global country-level CO_2 emissions	Long-run CO_2 emission forecasting and EKC analysis	LR, RF, SVR, XGB, NN	SHAP (Global + Local)	MAE, MSE, RMSE, R^2	Global + Local	Global country-level panel data (GDP, population, and CO_2 emissions)	Black-box: RF, XGB, SVR, ANN; Interpretable: LR	EKC pattern and country-specific effects identified	Forecast uncertainty, particularly for low-income countries, and limited predictors	Incorporate renewable energy, policy interventions, environmental regulations, and industry structure
Scopus	156	71	Chen et al. (2025)	Regional carbon management / Provincial emissions / Emission-reduction pathways	Provincial carbon emission prediction; scenario-based projections for 2030 and 2060	EBRB, CBRB, CBRB-IC, ARIMA, GM, XGBoost, BRB, DEA-EBRB, Bi-EBRB	SHAP (Global + Local); intrinsic rule-based interpretability	MAE, MSE, RMSE, MAPE, R; modeling time, inference time, rule count, antecedent attribute count	Global + Local	Provincial panel data; 30 Chinese provinces, 2004–2022; 11 candidate input indicators, reduced to five per region; statistical yearbooks and EDGAR	Black-box: XGBoost; interpretable statistical models: ARIMA, GM; rule-based models: BRB, EBRB, CBRB, DEA-EBRB, Bi-EBRB; interpretability-constrained rule-based model: CBRB-IC	Region-specific drivers: economic activity, fiscal revenue, transport, energy supply, and industrial indicators; five key inputs selected per region	Coarse regional grouping; limited indicators and geographic coverage; insufficient samples in data-scarce regions	City/county-level analysis; heterogeneous data integration; spatiotemporal models; transfer learning and synthetic data
Scopus	187	72	Mangla et al. (2024)	Energy economics / Energy risk management / GHG emissions	Total GHG emission prediction and identification of key emission drivers	RF, MLP, DNN	SHAP (Global + Local)	R^2, Adjusted R^2 (regression validation); Support, Confidence, Lift (association rules)	Global + Local	Multi-country numerical energy, emissions, climate, and socioeconomic indicator data	Black-box: RF, MLP, DNN	SHAP identified cumulative CO_2, cement emissions, and primary energy consumption as key predictors.	Limited regional real-time data and monitoring infrastructure; some DNN-derived feature importance findings lacked statistical validation	Real-time GHG monitoring; exploration of PCA with XAI; threshold-based emission classification evaluated using F1-score
Scopus	203	73	Crespo et al. (2021)	Regional and sectoral carbon emissions / Environmental policy	Total and sector-specific EU28 CO_2 emission prediction	NN/BP, NN/BP-BFE, NN/BP-RReliefF/BFE, Proposed NN/BP-RReliefF framework, GBM-BFE, RF-BFE, SVM-BFE	LIME (Local), RReliefF (Global feature ranking)	RMSE, MAPE	Global + Local	EU28 annual economic, energy, and total/sectoral CO_2 emissions data (1990–2017)	Black-box: NN/BP variants, GBM-BFE, RF-BFE, SVM-BFE	LIME quantified predictor contributions to individual annual CO_2 emission predictions	Restricted geographical and temporal coverage (EU28, 1990–2017)	Carbon emissions causality analysis; further development of ANN and XAI techniques
Scopus	54	74	Shi et al. (2026)	Energy / Urban sustainability / Carbon emissions	Energy-related CO_2 emission estimation and spatial reconstruction	LR, RFR, SVR, XGBoost	SHAP (Global + Interaction)	R^2, RMSE	Global + Interaction	Multi-source spatiotemporal, geospatial, remote sensing, and socioeconomic data (2000–2020)	Mixed (white-box: LR; black-box: RFR, SVR, XGBoost)	Feature importance, nonlinear thresholds, and interaction effects; NTL, GDP, and urban expansion identified as key emission drivers.	Annual data resolution, omitted explanatory variables, and limited spatial granularity.	Higher-frequency data, additional predictors, finer-scale analysis, and scenario-based emission projections.
Scopus	75	75	Qader and Albalooshi (2026)	Energy systems / Power-station CO_2 emissions	Power-station CO_2 emissions forecasting	XGBoost, Random Forest (RF), Long Short-Term Memory (LSTM), Support Vector Regression (SVR)	SHAP (Global + Local), LIME (Local)	RMSE, MAE, R^2, MAPE	Global + Local	Hourly plant-level operational, fuel, environmental, and temporal data (real-world and synthetic).	Mixed: ML + DL black-box models	Gross load and fuel consumption increase predicted CO_2 emissions, while higher unit efficiency reduces them.	Limited plant diversity, incomplete operational scenarios, and sensitivity to technological changes.	Broader plant validation, adaptive learning, physics-informed models, and advanced XAI techniques.
Scopus	78	76	Ramalingam et al. (2026)	Sustainable biofuel combustion / Engine emissions	Engine performance and emission prediction	XGBoost, AdaBoost, GBM	SHAP (Global + Local), Feature Importance Analysis (Global)	R^2, RMSE, MSE, MAE	Global + Local	Experimental engine data (165 observations)	Black-box: XGBoost, AdaBoost, GBM	Brake power and water concentration identified as key predictors	Small single-engine dataset; limited long-term validation	Lifecycle assessment, durability testing, and multi-cylinder validation

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Scopus	100	77	Sayghe et al. (2026)	Power systems / Energy forecasting / Smart grid	Load, electricity price, CO_2 emissions and renewable generation forecasting	ARIMA, Prophet, XGBoost, LGBM, LSTM, GRU, TCN, TFT, N-BEATS, Informer, Autoformer, PatchTST, TimesFM, Chronos, Moirai, Moirai-MoE, TimesFM-FT, Chronos-FT, Moirai-MoE-FT, GridFM	SHAP (Global), Attention Visualization (Global)	MAPE, sMAPE, RMSE, MAE, CRPS	Global	High-frequency, multi-regional spatiotemporal power grid time-series data	Black-box: XGBoost, LGBM, DL and foundation models; Statistical: ARIMA, Prophet	Key load and price drivers, daily patterns, and temporal dependencies identified	High training cost, extreme-event prediction challenges, and limited geographic generalizability	Cross-regional validation, online learning, and computationally efficient adaptation
Scopus	71	78	Sireesha & Sheik (2026)	Agriculture sector / Greenhouse gas (GHG) emissions	Prediction of CO_2, CH_4, and N_2O emissions (time-series regression	CNN, RNN, GRU, DFNN	SHAP (global & local), LIME, ICE, PDP	R^2, RMSE, MAE, VAF	Global + Local	Historical annual, multi-sector global GHG emissions time-series data (CO_2, CH_4, N_2O; 1850–2022; PIK database)	Black-box (deep learning models: CNN, RNN, GRU, DFNN)	CH_4 is the dominant contributor to predicted GHG emissions, followed by N_2O and CO_2, with sector-specific nonlinear effects.	Lack of independent external validation, potential overfitting, missing-data exclusions, and limited cross-sector generalizability.	Incorporate heterogeneous multi-source datasets, external validation, advanced temporal imputation, stronger regularization, and hybrid Transformer-based architectures.
Scopus	157	79	Jeaunita et al. (2025)	Precision Agriculture / Smart Farming	Carbon emission prediction / Anomaly detection	RF, LR, SVR, SVM, Shallow Autoencoder, Deep Autoencoder, Shallow FNN, FNN, LSTM, 1D CNN, Deep DNN, Bi-GRU, CNN, AlexNet, DDQGCN, Green ViT	SHAP (Global), LIME (Local)	R^2, MSE, RMSE, Accuracy, Precision, Recall, F1 Score	Global + Local	Multimodal agricultural sensor and drone imagery data	Black-box: Deep Autoencoder, FNN	SHAP-based feature importance for CO_2 prediction and anomaly detection	High computational demands and limited scalability	Edge computing integration for real-time processing
Scopus	45	80	Chai et al. (2026)	Enterprise carbon footprint tracking / Sustainable enterprises	Near-real-time Scope 1–2 emissions prediction and reduction optimization	Transformer, CNN, BiLSTM, Ensemble (Transformer + CNN + BiLSTM); Optimization: NSGA-III, PSO, SA	SHAP (Global + Local + Interaction), LIME (Local)	MAPE, RMSE, Pearson r, Emission reduction (%), ROI, NPV, Payback period	Global + Local	Multimodal enterprise data	Black-box DL: Transformer, CNN, BiLSTM, Ensemble (Transformer + CNN + BiLSTM); Metaheuristic optimization: NSGA-III, PSO, SA	Key emission drivers and reduction recommendations explained	18-month validation; IoT dependency and XAI computational costs.	Longitudinal validation, Scope 3 integration, edge optimization.
Scopus	24	81	Li et al. (2026)	Carbon markets / China's national ETS / Carbon allowance pricing	One-day-ahead carbon allowance return and closing-price prediction	RW, ARIMA, Ridge, LASSO, EN, KNN, SVR, RF, HGB, MLP, Ensemble	SHAP (Global + Local), PFI (Global), PDP (Global)	RMSE, MAE, MAPE, DA	Global + Local	Daily CEA market data; 1203 observations (2021–2026); 26 predictors	White-box: RW, ARIMA, Ridge, LASSO, EN; Black-box: KNN, SVR, RF, HGB, MLP; Mixed: Ensemble combining linear and black-box learners	Technical features dominated RF attribution; no stable forecast gains	Limited real-time data; proprietary inputs; short post-expansion period	Higher-frequency data; improved real-time inputs; longer validation period
Scopus	25	82	Ma et al. (2026)	Carbon markets / Carbon pricing / Multiscale driver analysis	One-day-ahead carbon price forecasting in China's Hubei carbon market	GRU, LSTM, ELM; EMD–GRU, EMD–LSTM, EMD–ELM; CEEMDAN–GRU, CEEMDAN–LSTM, CEEMDAN–ELM; CEEMDAN–SE–GRU, CEEMDAN–SE–LSTM, CEEMDAN–SE–ELM; VC ensemble, VMAPE ensemble, DMSFE ensemble, Mean ensemble, CS ensemble	SHAP (Global + Local)	MSE, MAE, RMSE, MAPE, R^2	Global + Local	Daily Hubei carbon-price and predictor data from Wind; December 2016–November 2024; 12 selected drivers	Black-box: GRU, LSTM, ELM, their decomposition/reconstruction hybrids, and forecast ensembles	Scale-specific contributions: sentiment and macroeconomic indicators in the short term; industrial costs and global markets at longer scales	Single train–test split; limited policy, unstructured, and firm-level data; one-way transmission analysis	Walk-forward validation; policy and sentiment text integration; firm-level data; feedback analysis

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Scopus	44	83	Sun et al. (2026)	Carbon markets / Regional ETS / Carbon pricing	Carbon price forecasting in Shanghai and Hubei across multiple horizons	CaStNet, CEEMDAN–SE–VMD + LSTM–Transformer, ALA–LSTM–Transformer, Informer, DLinear, LSTM–Transformer, TCN, CNN–LSTM, Transformer, GRU, XGBoost, SVR, ARIMA	SHAP (Global + Local + Interaction), LIME (Local)	RMSE, MAE, MAPE, R^2	Global + Local + Interaction	Daily Shanghai and Hubei market data through October 2025; over 2800 trading days per market; carbon, energy, quota, and financial variables	Interpretable linear models: ARIMA, DLinear; Black-box: CaStNet, CEEMDAN–SE–VMD + LSTM–Transformer, ALA–LSTM–Transformer, Informer, LSTM–Transformer, TCN, CNN–LSTM, Transformer, GRU, XGBoost, SVR	Historical-price contributions dominated short horizons; energy fundamentals gained importance at longer horizons; nonlinear energy-price thresholds were identified	Daily data; validation limited to Chinese markets; single-source Granger screening	Intraday and international validation; multidimensional state-driven attention; joint causal-source screening
Scopus	83	84	He et al. (2026)	Carbon markets / Carbon pricing / Risk monitoring	One-, five-, and ten-step-ahead carbon price forecasting	SVR, LSTM, GRU, BiGRU, TCN; CEEMDAN–CNN–SVR, CEEMDAN–CNN–LSTM, CEEMDAN–CNN–GRU, CEEMDAN–CNN–BiGRU, CEEMDAN–CNN–TCN; CEEMDAN–SVR–Attention, CEEMDAN–LSTM–Attention, CEEMDAN–GRU–Attention, CEEMDAN–BiGRU–Attention, CEEMDAN–TCN–Attention; CEEMDAN–CNN–SVR–Attention, CEEMDAN–CNN–LSTM–Attention, CEEMDAN–CNN–GRU–Attention,	SHAP (Global + Local), LIME (Local)	MAE, RMSE, MAPE, R^2	Global + Local	Daily EU and eight Chinese pilot-market prices; January 2014–August 2024; energy, financial, policy, sentiment, and environmental indicators	Black-box: SVR, LSTM, GRU, BiGRU, TCN, and their decomposition/CNN/attention hybrid	Energy, financial, policy, sentiment, and environmental contributions varied across markets; nonlinear thresholds were identified	Limited national-ETS representation; firm behavior and cross-market spillovers not explicitly modeled	Cross-regional interaction modeling; adaptive model updating; broader market validation
Scopus	171	85	Sayed et al. (2024)	Carbon markets / Carbon pricing / Investment decision support	Carbon price prediction from historical transaction features	MLSO–LSTM, LSTM, MSSA–LSTM, MWOA–LSTM, MAVOA–LSTM, MGWO–LSTM, MDE–LSTM, RF, GB	SHAP (Global + Local)	RMSE, R^2, MAE, MAPE, MSE	Global + Local	Historical transaction data reported for April 2005–April 2023; additional Beijing and Hubei ETS datasets; price, volume, change, and calendar features	Black-box: MLSO–LSTM, LSTM, MSSA–LSTM, MWOA–LSTM, MAVOA–LSTM, MGWO–LSTM, MDE–LSTM, RF, GB	Low, high, and opening prices dominated predictions; calendar features contributed little	Limited economic and policy inputs; random data splitting; unclear forecast-time feature availability	Policy, economic, and energy variables; alternative DL architectures; more complex datasets
Scopus	183	86	Deng et al. (2024)	Carbon markets / Carbon pricing / Directional forecasting	One-, five-, and ten-step-ahead carbon price direction classification in Hubei	MARS, SVM, RF, ELM, XGBoost, LSTM, LightGBM, CEEMDAN–LightGBM–S, CEEMDAN–LightGBM, CEEMDAN–TFS–LightGBM, CEEMDAN–BOA–LightGBM, CEEMDAN–TFS–BOA–LightGBM	SHAP (Global + Local)	AUC, Accuracy, Precision, Recall, F1 score	Global + Local	Daily Hubei carbon prices and external indicators; 1598 observations, January 2015–December 2021	Interpretable: MARS; Black-box: SVM, RF, ELM, XGBoost, LSTM, LightGBM, and all CEEMDAN–LightGBM variants	Historical-price IMFs dominated predictions; HC_IMF1, HC_IMF3, and HC_IMF4 were most influential at one-, five-, and ten-step horizons	Single-market validation; potentially incomplete influencing indicators; increased computational cost	Additional influencing factors; validation in other carbon markets